1 Features list

Categorized Features

This section shows the final list of features that have been used for predicting burned area and fire occurrence.

```
id_encoder
rhum_mean
Past_risk
fwi_max
cluster_encoder
isi_mean
rhum_max
bui_max
ffmc_mean
ffmc_max
dayofyear
sum_rain_last_7_days_mean
nesterov_mean
NDSI_min
calendar_sum
calendar_min
prec24h_max
angstroem_mean
sum_rain_last_7_days_max
munger_mean
dc_max
month
calendar_max
nesterov_max
rhum16_mean
precipitationIndexN5_max
precipitationIndexN5_mean
munger_max
angstroem_max
Pin maritime_max
precipitationIndexN3_max
foret_encoder_mean
Past_burnedarea
rhum_min
days_since_rain_max
temp_max
rhum16_max
Road_mean
```

temp16_max
fwi_min
NDSI_mean
sum_rain_last_7_days_min
prec24h16_max
dailySeverityRating_min
ffmc_min
bui_min
days_since_rain_min
Corine_vegetation_mean
rhum16_min
foret_encoder_max
Corine_vegetation_max
Chênes sempervirents_max
holidays
prcp_max
wdir16_mean
precipitationIndexN5_min
NDMI_min
munger_min
nesterov_min
Corine_grass_mean
wspd16_min
angstroem_min
Corine_urban_max
Mélèze_max
wdir_max
temp16_min
dc_min
dwpt_max
wdir_mean
population_mean
Pins mélangés_mean
Corine_littoral_max
Peuplier_max
dwpt16_max
NC_mean
NDMI_max
Mélèze_mean
Châtaignier_max
Pin laricio, pin noir_max
dwpt_min
NDSI_max
dwpt16_min
Corine_agricultural_mean
NDWI_max

Pins mélangés_max
wdir_min
NDMI_mean
Corine_water_mean
Pin d'Alep_max
Corine_grass_max
elevation_max
Corine_transport_mean
Conifères_mean
elevation_mean
wspd_max
NC_max
Pin autre_max
Peuplier_mean
prcp16_max
Douglas_mean
corine_encoder_mean
precipitationIndexN3_min
Feuillus_max
NR_max
Corine_rock_max
holidaysBorder
population_max
NDVI_mean
Pin à crochets, pin cembro_max
Sapin, épicéa_max
Conifères_max
NDVI_max
Pin sylvestre_mean
Mixte_max
Mixte_mean
Robinier_max
Pin laricio, pin noir_mean
wspd_mean
Douglas_max
wdir16_min
Corine_forest_mean
Pin sylvestre_max
Robinier_mean
wdir16_max
Chênes décidus_max
Corine_transport_max
Corine_water_max
Chênes décidus_mean
elevation_min
kbdi_max

wspd16_mean
Feuillus_mean
Corine_forest_max
Hêtre_mean
wspd_min
Hêtre_max
wspd16_max
prec24h16_min
prec24h_min
Corine_agricultural_max
corine_encoder_min
foret_encoder_min
ramadan
Corine_moisture_max
dayofweek
snow24h_max
snow24h16_max
sum_snow_last_7_days_max
isweekend
population_min
sum_snow_last_7_days_min
sum_consecutive_rainfall_max
prcp16_min
prcp_min
Road_min
snow24h_min
confinement
Feuillus_min
kbdi_min
sum_consecutive_rainfall_min
snow24h16_min
couvrefeux
id_encoder
cluster_encoder
rhum_mean
fwi_max
Past_risk
Pin maritime_max
foret_encoder_mean
isi_mean
bui_max
foret_encoder_max
nesterov_mean
Chênes sempervirents_max
ffmc_max
elevation_min

nesterov_max
Corine_vegetation_mean
angstroem_mean
dc_max
Corine_littoral_max
rhum16_mean
Corine_vegetation_max
Pin d'Alep_max
calendar_min
Corine_grass_mean
Road_mean
angstroem_max
Pins mélangés_mean
rhum_max
NDSI_min
corine_encoder_mean
ffmc_mean
dayofyear
month
Pin autre_max
munger_mean
kbdi_max
precipitationIndexN5_max
NDMI_min
prcp_max
NC_mean
sum_rain_last_7_days_mean
Pin laricio, pin noir_mean
rhum_min
sum_rain_last_7_days_min
Conifères_mean
rhum16_max
sum_rain_last_7_days_max
precipitationIndexN5_mean
Douglas_max
ffmc_min
Pins mélangés_max
Douglas_mean
precipitationIndexN3_max
Corine_transport_mean
Corine_rock_max
calendar_sum
days_since_rain_max
NDWI_max
dwpt16_max
Feuillus_mean

Conifères_max
Pin laricio, pin noir_max
NDSI_max
prec24h_max
NDMI_max
temp_max
Corine_forest_max
munger_max
NR_max
elevation_mean
Corine_grass_max
wspd16_mean
Chênes décidus_max
NDSI_mean
Past_burnedarea
temp16_max
Mixte_mean
rhum16_min
Pin sylvestre_max
angstroem_min
Corine_forest_mean
Peuplier_max
Mélèze_mean
Mélèze_max
population_mean
Sapin, épicéa_max
Peuplier_mean
wspd_mean
NC_max
dwpt_max
Hêtre_max
prcp16_max
Hêtre_mean
wdir16_mean
wdir16_max
Feuillus_max
wspd16_max
Robinier_max
precipitationIndexN5_min
nesterov_min
wspd_max
Corine_transport_max
munger_min
Corine_agricultural_mean
NDVI_max
bui_min

precipitationIndexN3_min
Châtaignier_max
calendar_max
Chênes décidus_mean
fwi_min
dailySeverityRating_min
days_since_rain_min
NDMI_mean
foret_encoder_min
Corine_water_max
Corine_moisture_max
NDVI_mean
Mixte_max
dc_min
Corine_water_mean
Corine_agricultural_max
prec24h16_min
wdir_max
corine_encoder_min
wspd_min
Corine_urban_max
prec24h_min
population_max
Pin à crochets, pin cembro_max
elevation_max
temp16_min
Pin sylvestre_mean
wdir_mean
sum_snow_last_7_days_max
prec24h16_max
wdir16_min
Robinier_mean
wspd16_min
dwpt_min
wdir_min
isweekend
dayofweek
snow24h_max
dwpt16_min
holidays
snow24h16_max
sum_consecutive_rainfall_max
holidaysBorder
ramadan
prcp16_min
population_min

prcp_min
Feuillus_min
sum_snow_last_7_days_min
confinement
Road_min
snow24h_min
sum_consecutive_rainfall_min
snow24h16_min
kbdi_min
couvrefeux
rhum_mean
rhum_max
NDSI_min
id_encoder
sum_rain_last_7_days_max
prec24h_max
Past_risk
fwi_max
sum_rain_last_7_days_mean
calendar_min
ffmc_mean
dayofyear
month
bui_max
isi_mean
precipitationIndexN3_max
Road_mean
calendar_sum
precipitationIndexN5_max
cluster_encoder
NDMI_min
nesterov_mean
angstroem_mean
dc_max
Pin maritime_max
rhum16_mean
precipitationIndexN5_mean
angstroem_max
nesterov_max
elevation_min
munger_mean
NDSI_mean
calendar_max
ffmc_max
dwpt16_max
prec24h16_max

Châtaignier_max
rhum16_max
Corine_grass_mean
Past_burnedarea
rhum_min
population_mean
NDSI_max
prcp_max
foret_encoder_mean
Corine_littoral_max
NDMI_max
fwi_min
temp_max
Corine_transport_mean
Pin d'Alep_max
ffmc_min
Corine_urban_max
corine_encoder_mean
days_since_rain_max
Peuplier_max
wspd16_min
dailySeverityRating_min
NDMI_mean
Chênes sempervirents_max
wdir_max
Pins mélangés_mean
wdir_mean
foret_encoder_max
dwpt_max
elevation_mean
dwpt_min
wspd_mean
days_since_rain_min
munger_max
Chênes décidus_max
Corine_agricultural_mean
wspd_max
dwpt16_min
bui_min
kbdi_max
Feuillus_max
sum_rain_last_7_days_min
Pins mélangés_max
munger_min
dc_min
wspd16_max

```
Mélèze_max
Feuillus_mean
Douglas_mean
Mélèze_mean
temp16_min
Peuplier_mean
temp16_max
Corine_grass_max
precipitationIndexN5_min
Douglas_max
wspd16_mean
Corine_forest_max
Pin autre_max
NDWI_max
wdir16_min
angstroem_min
population_max
holidays
Conifères_max
Corine_vegetation_max
Chênes décidus_mean
Corine_moisture_max
Pin sylvestre_mean
Corine_water_mean
Corine_vegetation_mean
Corine_forest_mean
nesterov_min
NC_max
elevation_max
Mixte_max
dayofweek
NC_mean
Pin laricio, pin noir_mean
Mixte_mean
NDVI_mean
NDVI_max
prcp16_max
Corine_agricultural_max
Corine_rock_max
wdir16_mean
Conifères_mean
wspd_min
NR_max
Robinier_max
Corine_transport_max
Pin sylvestre_max
```

Pin laricio, pin noir_max
rhum16_min
Robinier_mean
Hêtre_max
Corine_water_max
wdir_min
Sapin, épicéa_max
Pin à crochets, pin cembro_max
corine_encoder_min
wdir16_max
Hêtre_mean
foret_encoder_min
ramadan
precipitationIndexN3_min
isweekend
holidaysBorder
prec24h16_min
sum_snow_last_7_days_max
snow24h_max
prec24h_min
snow24h16_max
population_min
prcp16_min
sum_snow_last_7_days_min
confinement
prcp_min
sum_consecutive_rainfall_max
snow24h_min
Road_min
Feuillus_min
snow24h16_min
sum_consecutive_rainfall_min
kbdi_min
couvrefeux
id_encoder
rhum_mean
cluster_encoder
rhum_max
fwi_max
Past_risk
isi_mean
ffmc_mean
dc_max
Pin maritime_max
rhum16_mean
month

dayofyear
angstroem_mean
ffmc_max
calendar_min
angstroem_max
rhum16_max
calendar_sum
NDSI_min
precipitationIndexN5_max
sum_rain_last_7_days_mean
bui_max
precipitationIndexN5_mean
temp16_max
Road_mean
calendar_max
rhum16_min
rhum_min
temp_max
precipitationIndexN3_max
foret_encoder_mean
foret_encoder_max
prec24h_max
elevation_min
Past_burnedarea
NDMI_min
NDVI_max
angstroem_min
Corine_grass_mean
sum_rain_last_7_days_max
wdir_max
Chênes sempervirents_max
wdir16_mean
munger_max
Peuplier_mean
Peuplier_max
munger_mean
NDMI_mean
Pins mélangés_mean
dailySeverityRating_min
fwi_min
prec24h16_max
Pin sylvestre_max
nesterov_mean
Corine_vegetation_mean
NDSI_mean
wdir_mean

NC_mean
Corine_forest_mean
ffmc_min
Douglas_mean
corine_encoder_mean
Douglas_max
Pin laricio, pin noir_max
Mixte_mean
nesterov_max
Pin laricio, pin noir_mean
wspd_max
dwpt16_max
nesterov_min
wspd16_max
NDMI_max
precipitationIndexN5_min
wdir16_max
temp16_min
Pin d'Alep_max
dwpt_max
Corine_grass_max
sum_rain_last_7_days_min
days_since_rain_min
Feuillus_mean
wdir_min
Corine_vegetation_max
foret_encoder_min
kbdi_max
prcp_max
Chênes décidus_max
NDSI_max
NR_max
NC_max
Conifères_mean
Pin autre_max
Hêtre_max
Corine_transport_mean
Mélèze_max
Pins mélangés_max
Mélèze_mean
Corine_littoral_max
Châtaignier_max
munger_min
Pin à crochets, pin cembro_max
Sapin, épicéa_max
Corine_urban_max

population_mean
days_since_rain_max
wspd16_mean
Corine_rock_max
bui_min
Pin sylvestre_mean
Conifères_max
Corine_water_mean
Hêtre_mean
Corine_water_max
Robinier_max
wdir16_min
wspd_mean
dwpt16_min
Feuillus_max
dc_min
dayofweek
Corine_transport_max
NDWI_max
dwpt_min
Mixte_max
wspd16_min
wspd_min
elevation_mean
NDVI_mean
population_max
elevation_max
Robinier_mean
Chênes décidus_mean
Corine_forest_max
Corine_agricultural_mean
Corine_moisture_max
holidays
precipitationIndexN3_min
holidaysBorder
sum_snow_last_7_days_max
Corine_agricultural_max
ramadan
isweekend
prec24h16_min
snow24h_max
prcp16_max
corine_encoder_min
snow24h16_max
prec24h_min
sum_consecutive_rainfall_max

prcp16_min
population_min
prcp_min
sum_snow_last_7_days_min
confinement
Feuillus_min
Road_min
snow24h_min
sum_consecutive_rainfall_min
snow24h16_min
kbdi_min
couvrefeux
id_encoder
cluster_encoder
NDSI_min
rhum_mean
fwi_max
Past_risk
calendar_sum
ffmc_max
munger_max
munger_mean
calendar_max
rhum_max
bui_max
isi_mean
sum_rain_last_7_days_mean
ffmc_mean
days_since_rain_max
dc_max
dayofyear
Pin maritime_max
Past_burnedarea
nesterov_mean
rhum16_mean
sum_rain_last_7_days_max
bui_min
nesterov_max
month
ffmc_min
calendar_min
precipitationIndexN5_mean
angstroem_mean
kbdi_max
NDMI_min
prec24h_max

fwi_min
precipitationIndexN5_max
days_since_rain_min
Pins mélangés_max
dwpt_max
holidays
temp_max
foret_encoder_max
NDSI_max
Pins mélangés_mean
Pin laricio, pin noir_mean
angstroem_max
NDSI_mean
munger_min
temp16_min
nesterov_min
temp16_max
dailySeverityRating_min
NDVI_max
Pin laricio, pin noir_max
precipitationIndexN3_max
NR_max
Mélèze_max
dc_min
NDWI_max
Châtaignier_max
Mixte_max
rhum16_max
wdir_max
Feuillus_mean
Sapin, épicéa_max
Robinier_max
elevation_max
rhum_min
Chênes sempervirents_max
Mélèze_mean
foret_encoder_mean
dwpt16_max
NDVI_mean
wdir16_mean
Corine_water_mean
Chênes décidus_mean
Feuillus_max
prcp_max
elevation_mean
Corine_grass_mean

Chênes décidus_max
Hêtre_max
Corine_water_max
wspd16_max
wdir16_min
NDMI_mean
Peuplier_max
Hêtre_mean
Corine_forest_mean
Corine_forest_max
Road_mean
wspd16_min
NDMI_max
elevation_min
Corine_grass_max
wspd_mean
Pin sylvestre_mean
Corine_vegetation_max
Robinier_mean
rhum16_min
Pin à crochets, pin cembro_max
Corine_agricultural_mean
wspd_min
angstroem_min
Corine_transport_max
Mixte_mean
Douglas_mean
Douglas_max
NC_mean
wspd16_mean
population_max
Pin sylvestre_max
NC_max
wspd_max
population_mean
wdir_mean
Corine_littoral_max
Conifères_max
corine_encoder_mean
Corine_urban_max
prcp16_max
wdir16_max
holidaysBorder
Peuplier_mean
dwpt_min
prec24h16_max

```
dayofweek
Pin autre_max
precipitationIndexN3_min
sum_snow_last_7_days_max
wdir_min
Corine_vegetation_mean
dwpt16_min
Corine_agricultural_max
sum_rain_last_7_days_min
Corine_rock_max
Corine_transport_mean
Conifères_mean
prec24h16_min
Pin d'Alep_max
Corine_moisture_max
precipitationIndexN5_min
corine_encoder_min
snow24h_max
foret_encoder_min
prec24h_min
snow24h16_max
isweekend
ramadan
population_min
sum_consecutive_rainfall_max
prcp16_min
prcp_min
sum_snow_last_7_days_min
Feuillus_min
confinement
Road_min
snow24h_min
sum_consecutive_rainfall_min
snow24h16_min
kbdi_min
couvrefeux
```

2 Tree based configuration

Table 1 presents the configuration of tree based models.

Table 1: **Configuration of each tree based model**

Parameters	**XGBoost**	**Catboost**
Early Stopping Rounds	15	15
Learning Rate	0.001	0.001
Max Depth/Num Leaves	6	4
Min Child Weight	1.0	-
Max Delta Step	1.0	-
Subsample	0.5	0.7
Col Sample Bytree	0.7	0.6
Reg Lambda	1.7	1
Reg Alpha	0.7	0.27
N Estimators	10000	10000
Tree Method	hist	-

3 Deep learning layers configuration

This section present the layers parameters for each deep learning model used in this study.

3.1 MLP

Table 2 shows the configuration of the MLP models.

Table 2: Layer-wise architecture of the MLP model

Layer	**Configuration / Dimensions**
Linear 1	Linear(179 $\rightarrow$ 179), ReLU
Linear 2	Linear(179 $\rightarrow$ 179), ReLU
Linear 3	Linear(179 $\rightarrow$ 64), ReLU
Linear 4	Linear(64 $\rightarrow$ 5)

3.2 GraphCastGRY

Table 3 shows the configuration of the GraphCast model. Edges channels is set to 4 as in the original paper containing the x, y, z positions of the mesh node and the length of the edge. Similarly, the input mesh data is the 3D position of the node.

3.3 GRU

Table 4 shows the configuration of the GRU model.

Table 3: Layer-wise architecture of the `GraphCastGRU` model

Layer	Configuration / Dimensions
GRU	GRU(num_layers=2, input_size=`in_channels`, hidden_size=128)
BatchNorm	BatchNorm1d(128)
Dropout	Dropout(0.03)
GraphCastNet	Message Passing (6 layers), hidden dim = 128
Linear 1	Linear(128 $\rightarrow$ 256)
Linear 2	Linear(256 $\rightarrow$ 64)
Output Layer	Linear(64 $\rightarrow$ `out_channels`)
Activation (internal)	ReLU
Activation (output)	Softmax (for classification)

Table 4: Layer-wise architecture of the GRU model

Layer	Configuration / Dimensions
GRU	2 layers, hidden size 179, ReLU, dropout 0.03
Norm 1	BatchNorm
Dropout	0.03
Linear 1	Linear(179 $\rightarrow$ 256), ReLU
Linear 2	Linear(256 $\rightarrow$ 64), ReLU
Linear 3	Linear(64 $\rightarrow$ 5)

3.4 LSTM

Table 5 shows the configuration of the LSTM model.

Table 5: Layer-wise architecture of the LSTM model

Layer	Configuration / Dimensions
LSTM	1 layer, hidden size 64, ReLU
Norm 1	BatchNorm
Dropout	0.03
Linear 1	Linear(64 $\rightarrow$ 64), ReLU
Linear 2	Linear(64 $\rightarrow$ 64), ReLU
Linear 3	Linear(64 $\rightarrow$ 5)

3.5 DilatedCNN

Table 6 shows the configuration of the DilatedCNN model.

4 Features importance

This section features importance based on shap values for Catboost models. Figure 1 shows features importance of Burned area prediction of Catboost model.

Table 6: Layer-wise architecture of the DilatedCNN model (Dil = dilation)

Layer	**Configuration / Dimensions**
Conv 1	Conv1d(179 filters, dilation 1)
Conv 2	Conv1d(179 filters, dilation 3)
Conv 3	Conv1d(dilation 3)
Norm + Dropout	BatchNorm, dropout 0.03
Linear 1	Linear(179 $\rightarrow$ 64)
Linear 2	Linear(64 $\rightarrow$ 64)
Linear 3	Linear(64 $\rightarrow$ 5)

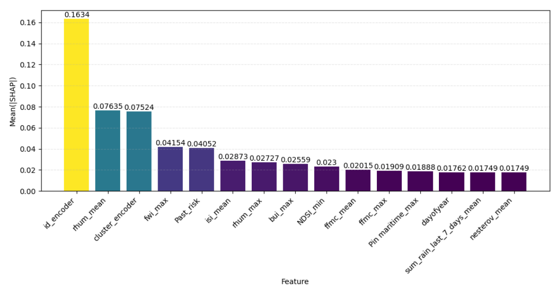

Figure 1: Top 15 shap values computed on multi-classification Catboost model for Burned area prediction. ID encoder correspond to Departmment ID